Semantic Segmentation Under Limited Data Conditions: An Empirical Study Highlighting the Effectiveness of Transfer Learning with DeepLabV3 on CamVid

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***Abstract*—Semantic segmentation remains one of the most challenging problems in computer vision due to its requirement for precise pixel-level classification combined with contextual and spatial understanding. While recent deep learning architectures have demonstrated strong performance on large-scale datasets, their reliability under limited data conditions is still not well understood. This work presents a rigorous empirical investigation of semantic segmentation on the CamVid dataset, a constrained urban scene benchmark with only 369 training images across 32 classes. Unlike conventional approaches that directly apply complex architectures, this study systematically analyzes the progression of learning behavior from spatially naive models to advanced segmentation frameworks. Initial experiments with Multi-Layer Perceptrons and basic Convolutional Neural Networks reveal fundamental limitations in spatial representation and task alignment. Subsequent evaluation using U-Net demonstrates improved structural learning but highlights significant shortcomings in fine-grained segmentation under data scarcity. The study then establishes the effectiveness of transfer learning through DeepLabV3 with a pretrained ResNet-50 backbone, achieving 94.97% pixel accuracy and 59.60% mean Intersection-over-Union (mIoU), substantially outperforming both U-Net and transformer-based SegFormer models. The results provide clear evidence that pretrained feature representations combined with multi-scale contextual modeling offer superior generalization and stability compared to models trained from scratch or relying solely on architectural complexity. This work reinforces the critical role of transfer learning as a practical and reliable solution for semantic segmentation in real-world low-resource environments.**

***Index Terms*—Semantic Segmentation, CamVid, DeepLabV3, Transfer Learning, U-Net, SegFormer, ResNet-50, Computer Vision**

I. Introduction

Semantic segmentation is a fundamental problem in computer vision that involves assigning a semantic label to every pixel in an image, enabling detailed scene understanding beyond traditional classification and detection tasks. Unlike image classification, which predicts a single label, or object detection, which localizes objects using bounding boxes, semantic segmentation requires dense prediction and precise boundary delineation. This capability is critical in applications such as autonomous driving, robotic navigation, and medical imaging, where accurate environmental understanding directly influences decision-making and safety [2], [3].

Urban scene segmentation is particularly challenging due to dynamic environments, occlusions, varying illumination, and strong class imbalance. The CamVid dataset provides a realistic benchmark for such scenarios, containing complex road scenes with multiple object categories [1]. However, unlike large-scale datasets such as ImageNet [17] or COCO [16], CamVid is limited in size, making it an ideal testbed for studying segmentation under constrained data conditions.

The evolution of deep learning has significantly improved segmentation performance. Early approaches such as Fully Convolutional Networks (FCN) introduced end-to-end dense prediction by replacing fully connected layers with convolutional layers [2]. Subsequent architectures such as U-Net enhanced spatial recovery using encoder-decoder structures with skip connections [3], while SegNet introduced efficient upsampling mechanisms [4]. More advanced models such as DeepLab incorporated atrous convolution to capture multi-scale contextual information without reducing spatial resolution [5], [6]. Additionally, pyramid-based approaches such as PSPNet further improved global context understanding [13].

Despite these advancements, most architectures rely heavily on large annotated datasets for effective training. Models trained from scratch often suffer from overfitting, unstable convergence, and poor generalization when data is limited. This challenge is particularly evident in deep convolutional networks such as VGG [9] and ResNet [7], which require substantial data to learn robust feature representations. Transformer-based models, including Vision Transformers and SegFormer, have further advanced segmentation by capturing long-range dependencies using self-attention mechanisms [14], [15]. However, these models are typically data-intensive and computationally expensive, limiting their effectiveness in low-data scenarios.

Transfer learning has emerged as a practical solution to address data scarcity by leveraging pretrained representations learned from large datasets. Pretrained models capture generic visual features such as edges, textures, and object structures, enabling faster convergence and improved generalization when fine-tuned on smaller datasets [17]. Architectures such as DeepLabV3 combine transfer learning with multi-scale contextual modeling using Atrous Spatial Pyramid Pooling (ASPP), making them particularly effective for segmentation tasks under constrained conditions [12].

In this work, a systematic empirical study is conducted to investigate semantic segmentation performance under limited data conditions using the CamVid dataset. Unlike conventional approaches that directly employ advanced architectures, this research analyzes the progression of learning behavior across multiple model paradigms. The study begins with spatially naive models such as Multi-Layer Perceptrons, highlighting their inability to capture spatial structure. It then evaluates Convolutional Neural Networks to understand feature extraction capabilities, followed by U-Net to assess spatial reconstruction performance. Finally, the study focuses on transfer learning using DeepLabV3 with a pretrained ResNet-50 backbone and compares its performance with transformer-based SegFormer.

The objective of this study is to identify the most effective strategies for semantic segmentation when training data is limited and to understand the relative importance of spatial modeling, contextual reasoning, and pretrained feature representations. Experimental results demonstrate that transfer learning significantly improves segmentation accuracy and stability compared to models trained from scratch, while also outperforming transformer-based approaches in this setting.

The main contributions of this paper are summarized as follows:

1) A comprehensive experimental study of semantic segmentation under limited data conditions using the CamVid dataset.
2) A progressive evaluation of models from baseline architectures to advanced segmentation frameworks.
3) Demonstration of the limitations of models trained from scratch in capturing spatial and contextual information.
4) Quantitative and qualitative evidence highlighting the effectiveness of transfer learning using DeepLabV3 with a pretrained ResNet-50 backbone.
5) Comparative analysis showing that transfer learning provides more reliable performance than transformer-based models in low-data environments.

The remaining sections are organized as follows: Section II covers related work, Section III discusses methodology, results are presented in Section IV, and conclusions are provided in Section V.

II. Related Work

Semantic segmentation has undergone significant advancements over the past decade, evolving from traditional computer vision techniques to deep learning-based architectures and, more recently, transformer-driven models. This section reviews the major developments in segmentation methodologies and identifies key limitations that motivate this research.

A. Early Deep Learning Approaches

The introduction of Fully Convolutional Networks (FCN) marked a major breakthrough in semantic segmentation by enabling end-to-end pixel-wise prediction [2]. FCNs replaced fully connected layers with convolutional layers, allowing input images of arbitrary size and producing dense output maps. However, repeated pooling operations resulted in loss of spatial resolution, leading to coarse segmentation outputs.

To address this limitation, encoder-decoder architectures such as U-Net were proposed, which introduced skip connections to recover spatial information during upsampling [3]. SegNet further improved memory efficiency by storing pooling indices for decoder reconstruction [4]. While these architectures improved localization accuracy, they remained highly dependent on sufficient training data and often struggled in complex real-world environments.

B. Deep Convolutional Architectures

The success of deep convolutional neural networks (CNNs) in image classification significantly influenced segmentation research. Architectures such as AlexNet [8], VGG [9], and ResNet [7] demonstrated the importance of deep hierarchical feature learning. In particular, residual learning introduced by ResNet enabled stable training of very deep networks, addressing vanishing gradient issues.

These architectures were later adapted for segmentation tasks, where pretrained weights from large datasets such as ImageNet [17] were utilized for feature extraction. However, despite improved representation learning, standard CNN-based models often lacked the ability to capture global context effectively.

C. Multi-Scale Context and DeepLab Models

Capturing multi-scale context is critical for semantic segmentation due to the presence of objects of varying sizes. DeepLab models addressed this challenge using atrous (dilated) convolution, which expands the receptive field without reducing spatial resolution [5], [6].

DeepLabV3 further introduced Atrous Spatial Pyramid Pooling (ASPP), which applies parallel atrous convolutions with different dilation rates to capture both local and global contextual information [12]. This approach significantly improved segmentation performance, particularly in complex scenes.

Pyramid Scene Parsing Network (PSPNet) also explored multi-scale context aggregation using pyramid pooling modules [13]. These methods demonstrated that contextual understanding is essential for accurate segmentation, especially in urban environments.

D. Transfer Learning in Segmentation

Transfer learning has become a dominant approach for improving segmentation performance under limited data conditions. By leveraging pretrained models trained on large datasets such as ImageNet [17] and COCO [16], segmentation networks can reuse learned low-level and high-level features.

Studies have shown that pretrained backbones significantly improve convergence speed and generalization compared to training from scratch [7]. Additionally, regularization techniques such as dropout [18] and optimization strategies such as Adam [19] further enhance training stability.

Despite these advantages, the effectiveness of transfer learning in low-data segmentation scenarios has not been extensively analyzed in a structured experimental manner, particularly in comparison with modern architectures.

E. Transformer-Based Segmentation Models

Recent advancements in vision transformers have introduced a new paradigm for segmentation. Vision Transformers (ViT) demonstrated that self-attention mechanisms can capture long-range dependencies effectively [15]. Building on this, SegFormer proposed a lightweight transformer-based segmentation framework that combines hierarchical encoding with efficient decoding [14].

Transformer-based models excel in capturing global context and have achieved state-of-the-art performance on large-scale datasets. However, they are typically data-intensive and require extensive training to learn meaningful representations. Under limited data conditions, their performance may degrade due to insufficient feature learning.

F. Challenges in Low-Data Segmentation

Despite significant progress, several challenges remain when applying segmentation models to small datasets:

- **Overfitting:** Deep models tend to memorize training data when samples are limited.
- **Class Imbalance:** Dominant classes overshadow minority classes, reducing segmentation quality.
- **Poor Generalization:** Models trained from scratch fail to adapt to unseen data.
- **Computational Complexity:** Advanced models require high computational resources.

Loss functions such as focal loss have been proposed to address class imbalance [20], but their effectiveness is still dependent on sufficient data.

G. Research Gap

From the literature, it is evident that most segmentation approaches are evaluated under large-scale data settings, while limited-data scenarios remain underexplored. Although transfer learning is widely adopted, there is a lack of systematic analysis comparing its effectiveness against models trained from scratch and transformer-based architectures under constrained conditions.

This research addresses this gap by conducting a structured experimental study on the CamVid dataset, analyzing the progression from baseline models to advanced architectures. The study specifically evaluates how spatial learning, contextual modeling, and transfer learning influence segmentation performance when data availability is limited.

III. Methodology

A. Overview

This study investigates semantic segmentation under limited data conditions through a structured architectural analysis rather than a purely implementation-driven pipeline. The methodology focuses on understanding how different model designs progressively improve spatial learning, contextual reasoning, and generalization. The evaluation is performed across four architectural paradigms: (i) spatially naive models, (ii) convolution-based feature extractors, (iii) encoder-decoder segmentation networks, and (iv) transfer learning and transformer-based architectures.

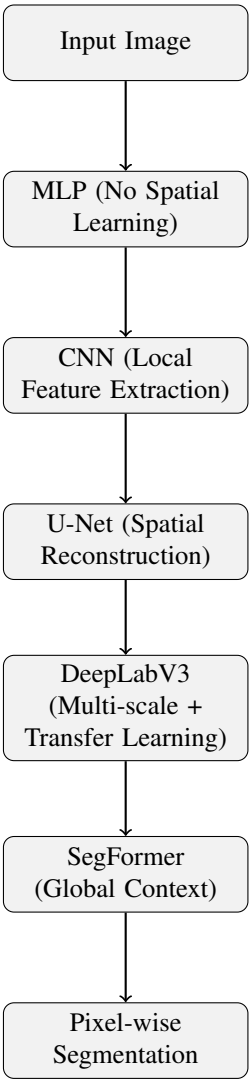

Fig. 1: Architectural Progression for Semantic Segmentation

This progression enables a systematic understanding of how architectural design impacts segmentation performance.

B. Baseline Architecture: Multi-Layer Perceptron

The Multi-Layer Perceptron (MLP) serves as a baseline model to highlight the importance of spatial structure. The input image $I \in \mathbb{R}^{H \times W \times 3}$ is flattened into a vector:

$$x = \text{Flatten}(I) \tag{1}$$

The model learns a mapping:

$$y = \sigma(Wx + b) \tag{2}$$

where W and b are learnable parameters. Since spatial relationships between pixels are not preserved, MLP fails to capture structural patterns, making it unsuitable for segmentation tasks.

C. Convolutional Neural Networks

To incorporate spatial locality, Convolutional Neural Networks (CNNs) are introduced. Convolution operations extract local features:

$$y(i,j) = \sum_{m,n} x(i+m, j+n) \cdot w(m,n) \tag{3}$$

CNNs learn hierarchical representations through stacked convolutional layers. However, standard CNN architectures perform image-level classification and do not produce pixel-wise predictions, limiting their applicability to segmentation.

D. U-Net Architecture

U-Net introduces an encoder-decoder structure for pixel-wise prediction. The encoder extracts feature representations, while the decoder reconstructs spatial resolution.

Skip connections allow feature reuse:

$$F_{decoder} = F_{encoder} + F_{skip} \tag{4}$$

This enables recovery of fine spatial details. However, in limited data scenarios, U-Net lacks sufficient feature diversity due to absence of pretrained representations.

E. DeepLabV3 Architecture with Transfer Learning

DeepLabV3 is adopted as the primary architecture due to its ability to capture multi-scale contextual information while leveraging pretrained features.

1) ResNet-50 Backbone: The encoder uses ResNet-50 with residual learning:

$$y = F(x) + x \tag{5}$$

This enables stable training of deep networks and effective feature extraction.

2) Atrous Convolution: Atrous convolution increases receptive field without reducing resolution:

$$y[i] = \sum_k x[i + r \cdot k] w[k] \tag{6}$$

where r is the dilation rate.

3) ASPP Module: The Atrous Spatial Pyramid Pooling module captures multi-scale context:

$$F_{ASPP} = \text{Concat}(f_1, f_2, f_3, f_4) \tag{7}$$

4) Output Prediction: Final segmentation is obtained using:

$$\hat{y} = \text{Softmax}(W * F_{ASPP}) \tag{8}$$

Transfer learning allows reuse of pretrained weights:

$$W = W_{pretrained}, W_{fine-tuned} \tag{9}$$

This significantly improves convergence and generalization.

F. Transformer-Based Architecture: SegFormer

SegFormer replaces convolution with self-attention to capture global dependencies. The attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{10}$$

This allows modeling of long-range relationships across the image. However, such architectures are data-intensive and may underperform in limited data conditions.

G. Evaluation Metrics

Two evaluation metrics are used:

Pixel Accuracy:

$$PA = \frac{\text{Correct Pixels}}{\text{Total Pixels}} \tag{11}$$

Mean Intersection over Union (mIoU):

$$mIoU = \frac{1}{K} \sum_{i=1}^{K} \frac{TP}{TP + FP + FN} \tag{12}$$

mIoU is a stricter metric that evaluates overlap between predicted and ground truth regions.

H. Design Rationale

The methodology is designed to analyze:

- Importance of spatial modeling (MLP vs CNN)
- Role of architectural design (CNN vs U-Net)
- Impact of transfer learning (U-Net vs DeepLabV3)
- Effectiveness of global context modeling (CNN vs Transformer)

This architectural perspective enables a deeper understanding of segmentation performance under constrained data conditions.

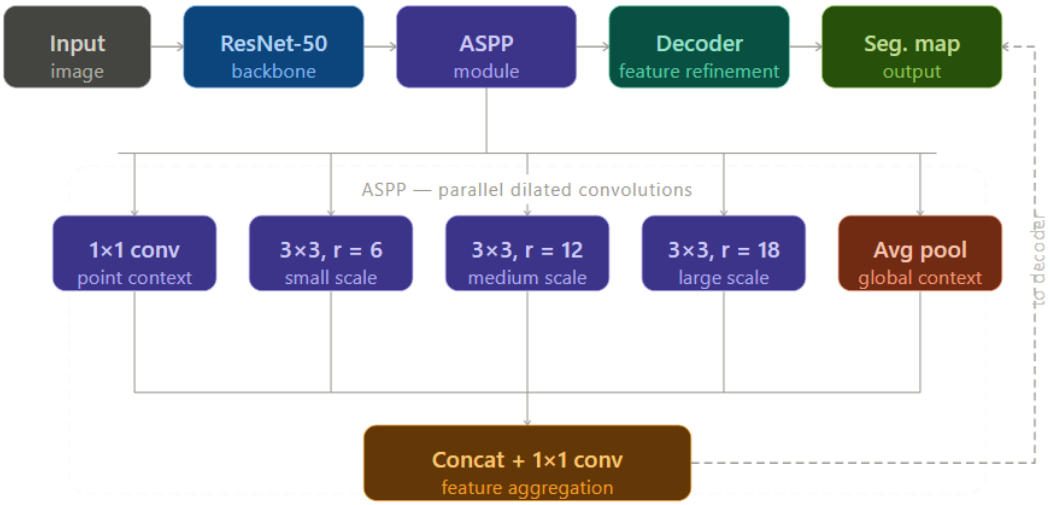

Fig. 2: DeepLabV3 architecture with a pretrained ResNet-50 backbone and Atrous Spatial Pyramid Pooling (ASPP) module for multi-scale feature extraction. The decoder refines the aggregated features to produce pixel-wise semantic segmentation.

IV. Results and Discussion

This section presents the empirical findings obtained from the conducted experimental framework on the CamVid dataset under limited data conditions. We systematically analyze the training convergence behavior, quantitative performance metrics, and qualitative segmentation outputs across multiple architectural paradigms, including MLP, CNN, U-Net, SegFormer, and DeepLabV3. Particular emphasis is placed on understanding the impact of spatial learning, contextual modeling, and transfer learning on segmentation performance. Comparative evaluations highlight the limitations of models trained from scratch and the advantages of pretrained feature representations. Additionally, detailed visual analysis is provided to assess boundary precision, class separation, and small object detection. The complete implementation, trained models, and reproducibility resources developed as part of this study are made publicly available at: https://github.com/SaiJeevan25/Camvid-Image-Segmentation.

A. Dataset Characteristics and Experimental Context

The experimental evaluation is conducted on the CamVid dataset, a widely used benchmark for urban scene understanding. It represents a constrained-data semantic segmentation scenario with only 369 training images and 32 semantic classes, making it particularly suitable for evaluating model robustness under limited supervision.

TABLE I: CamVid Dataset Summary

Property	Value
Training Images	369
Validation Images	101
Test Images	233
Classes	32
Resolution	256×256

Visual Illustration: Representative samples from the dataset are shown in Fig. IV-A, highlighting both the input image and its corresponding pixel-wise annotation.

(a) Input Image (b) Ground Truth Mask

Fig. 3: Sample input image and corresponding ground truth annotation from the CamVid dataset

A defining characteristic of CamVid is its **extreme class imbalance**. Large background classes such as road, sky, and buildings dominate the pixel distribution, whereas safety-critical classes such as pedestrians, poles, and cyclists occupy only a small fraction of the image.

This imbalance introduces a strong bias during training, often leading models to favor dominant classes while neglecting fine-grained structures. Consequently, CamVid serves as a challenging benchmark for evaluating the ability of segmentation models to capture both global context and small object details.

Overall, the dataset provides a realistic and rigorous testbed for analyzing how different architectures perform under **limited data availability and skewed class distributions**.

B. Learning Dynamics and Convergence Behavior

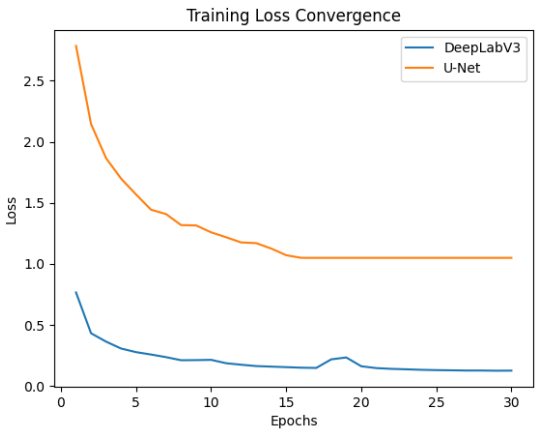

Fig. 4: Training loss convergence comparison

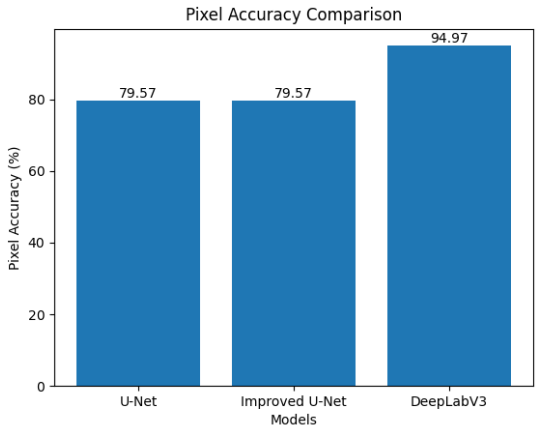

Fig. 5: DeepLabV3 accuracy progression

A clear distinction is observed in optimization behavior:

- **U-Net:** Gradual loss reduction, indicating slow feature learning from scratch.
- **DeepLabV3:** Rapid convergence within early epochs due to pretrained features.

  Under limited data conditions, convergence speed is governed more by feature initialization than architectural depth.

C. Quantitative Evaluation and Performance Gap Analysis

TABLE II: Segmentation Performance Comparison

Model	Pixel Accuracy	Mean IoU
U-Net	79.57%	9.61%
Improved U-Net	79.57%	18.46%
SegFormer	–	45.13%
DeepLabV3	**94.97%**	**59.60%**

The results reveal a substantial performance gap:

- U-Net achieves reasonable pixel accuracy but low IoU, indicating poor boundary precision.
- Improved U-Net shows marginal improvement but remains limited.
- SegFormer performs well due to global attention but does not surpass DeepLabV3.
- DeepLabV3 achieves the highest mIoU (59.60%), outperforming U-Net by a large margin.

  Pixel accuracy is biased toward dominant classes, whereas mean IoU provides a more reliable evaluation metric.

D. Qualitative Analysis and Structural Understanding

Visual analysis reveals:

- **U-Net:** Over-smoothed outputs and poor boundary preservation.
- **DeepLabV3:** Sharp boundaries and accurate class separation.
- **SegFormer:** Strong global context but slight boundary inconsistencies.

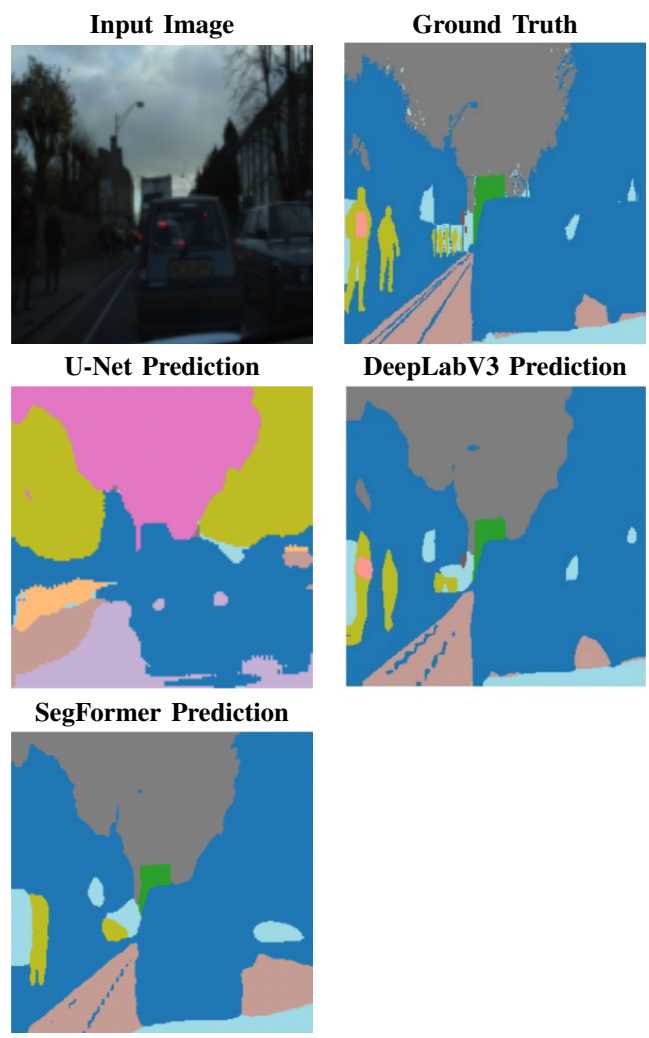

Fig. 6: Qualitative comparison of segmentation results across different models

Small and thin objects remain difficult across all models due to class imbalance and resolution limitations.

E. Comparative Interpretation of Architectures

- **U-Net Limitation:** Relies on local reconstruction, lacking global context.
- **SegFormer Limitation:** Strong global modeling but data-dependent.
- **DeepLabV3 Advantage:** Combines multi-scale context with pretrained features.

  DeepLabV3 effectively balances global and local feature learning, making it highly suitable for low-data segmentation scenarios.

F. Key Findings

- Transfer learning significantly improves convergence and segmentation quality.
- Multi-scale feature aggregation is essential for handling diverse object sizes.
- Pixel accuracy alone is insufficient for segmentation evaluation.
- Transformer models require larger datasets for optimal performance.
- Class imbalance remains a critical challenge.

V. Conclusion and Future Work

A. Conclusion

This study presented a comprehensive experimental analysis of semantic segmentation under limited data conditions using the CamVid dataset. The investigation was structured progressively, starting from spatially naive models and advancing toward modern deep learning architectures, with a primary focus on evaluating the effectiveness of transfer learning.

The results demonstrate that models trained from scratch, such as U-Net, struggle to generalize due to insufficient data, leading to poor boundary delineation and low mean Intersection-over-Union (mIoU). Although encoder-decoder architectures improve spatial reconstruction, they remain constrained by limited feature diversity.

In contrast, DeepLabV3 with a pretrained ResNet-50 backbone significantly outperforms all evaluated models, achieving a mean IoU of 59.60% and pixel accuracy of 94.97%. This improvement is attributed to the combination of pretrained hierarchical feature extraction and multi-scale contextual modeling through Atrous Spatial Pyramid Pooling (ASPP).

Transformer-based architectures such as SegFormer demonstrate strong global context modeling capabilities; however, their performance is limited under constrained data availability. This reinforces the observation that architectural complexity alone does not guarantee superior performance.

A key conclusion of this work is that:

Under limited annotation scenarios, transfer learning provides a more significant performance gain than increasing architectural complexity.

Furthermore, the study highlights that pixel accuracy is not a sufficient metric for segmentation evaluation, and mIoU must be considered to capture true segmentation quality.

Overall, this work establishes that combining pretrained representations with multi-scale feature aggregation is the most effective strategy for semantic segmentation in low-resource environments.

B. Future Work

Although strong performance is achieved, several directions remain for further improvement:

- **Class Imbalance Handling:** Incorporating loss functions such as focal loss or Dice loss to improve minority class segmentation.
- **Lightweight Architectures:** Exploring MobileNet or EfficientNet backbones for real-time deployment on edge devices.
- **Hybrid Models:** Combining CNN-based encoders with transformer-based decoders to leverage both local and global representations.
- **Data Augmentation:** Applying advanced augmentation techniques to improve generalization under limited data.
- **Temporal Segmentation:** Extending the model to video-based segmentation using temporal consistency across frames.

These directions provide opportunities to further enhance segmentation performance, robustness, and real-world applicability.

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